Name: _____

DNA Extraction – Inquiry Task GEHBY

You are completing a DNA extraction and the questions on this sheet will be assessed. The amount of marks given for each question is in brackets. You will complete the virtual extraction on Tuesday and the strawberry extraction on Wednesday.

Website 1 is the virtual extraction. Website 2 is an extra fact sheet that you may find useful.

*Note: the questions marked with an asterisk are not necessarily given in the simulation and will require a little extra thought/research. If you are having trouble please ask me AFTER you have tried to find the answer.

(1) https://learn.genetics.utah.edu/content/labs/extraction/
(2) http://www.explorecuriocity.org/portals/2/themes/biotechnology/DNA-Extraction-Backgrounder.pdf

Questions from the virtual extraction:

1. Give three examples of why we might need to extract DNA from humans. (3)

2. Why would a blood sample not be as ideal for DNA extraction? (2)

3. *What type of tissue are buccal (cheek) cells? What makes them suited to the purpose of DNA extraction? (2)

4. What are the 4 steps in DNA extraction? (4)

5. What are the two main ingredients in the lysis solution and what do they do? (4)

6. *Why is the sample put in a warm water bath after the lysis solution is added? (2)

7. Describe the 2 methods of physical separation involved in separating the DNA from the proteins and other debris. (2)

8. What does adding the isopropyl alcohol do and why? (2)

9. *Sometimes the DNA can become contaminated with other DNA not from the target organism. List 2 possible contaminants and describe where in the process they might occur. (4)

10. *In preparation for the strawberry lab, research why strawberries are often used for DNA extractions.

    Define polyploidy (1):

    Explain why strawberries are used (2):

Strawberry Lab

11. In the Strawberry DNA extraction we used mechanical and chemical lysis. Describe both of these processes in the lab. (2)

12. Propose a reason why the human sample did not use mechanical lysis but the strawberry lab does (1)

13. What step/s in the virtual lab are analogous (similar) to the filtering of the strawberry pulp? (2)

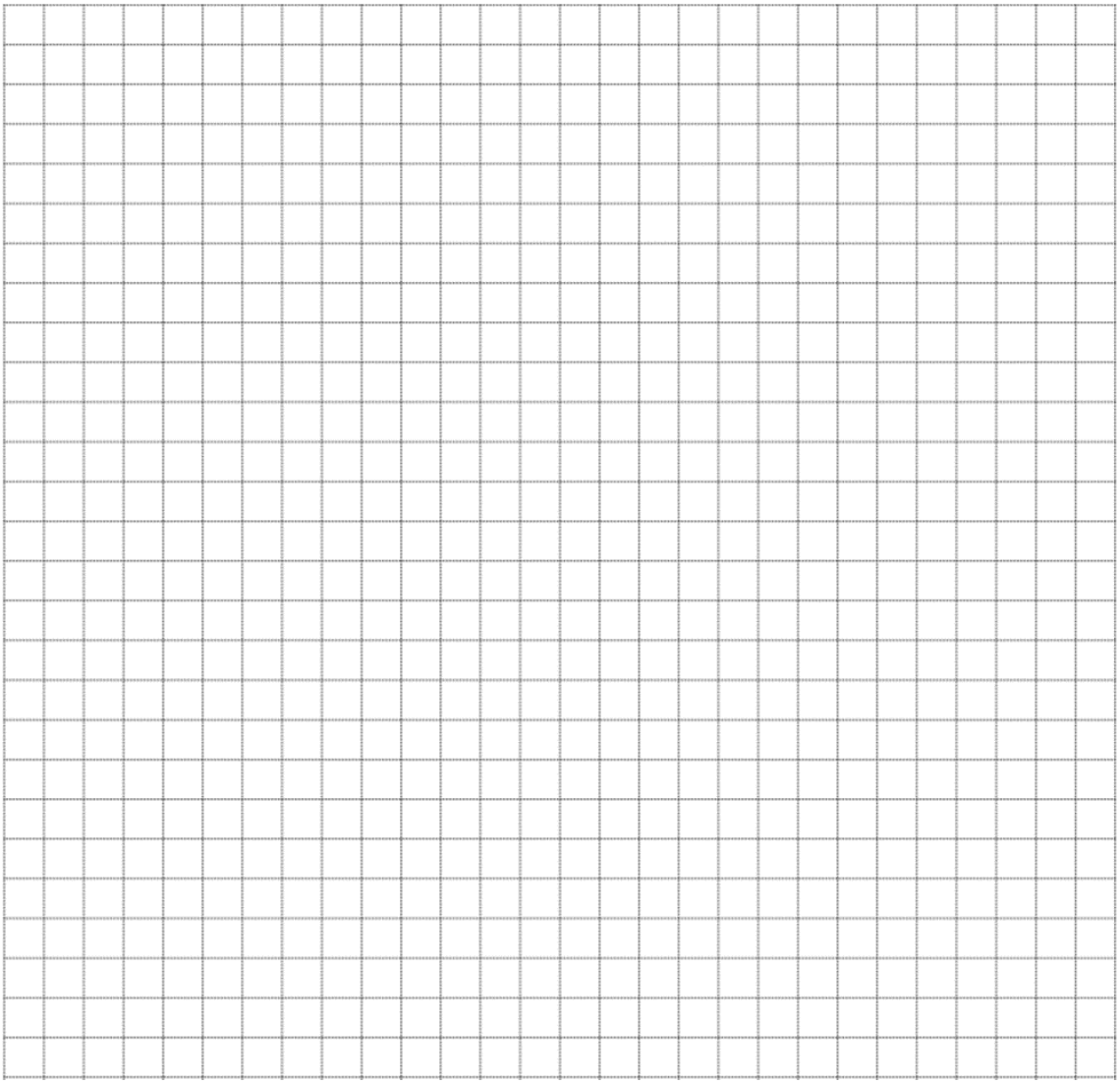
14. What is left behind in the filtrate? (2)

Graphing task:

Type of Fruit	Amount of DNA extracted (grams)			Average
	Trial 1	Trial 2	Trial 3	
Banana	5.6	6.2	11.1	
Kiwifruit	7.3	7.9	8.5	
Strawberry	11.4	12.3	10.7	

Create a graph for this data (5)

- Title relates dependent and independent variables
- Appropriate labels and units
- Type of graph appropriate
- Appropriate size and scale
- Calculating averages correctly and graphing the appropriate data



Write a hypothesis which the researcher may have written. (2)

What was the independent variable? (1)

What was the dependent variable? (1)

List 5 things that could have been controlled in this experiment. (5)

What might you infer from this data regarding the content of DNA extracted and polyploidy? (1)

Mark: /35